

Business Analyst Intern Technical Assessment - Set B

Executive Summary

This report analyzes 12,000 hotel booking records and addresses all four sections of the assessment: data quality auditing, a CFO-focused discount erosion case study, SQL schema design and analytics, and a holiday-proximity demand tagging mini-project using a public API. The analysis identifies OTA as the highest discount-intensity channel, finds limited behavioral benefit from coupons, estimates recoverable margin of approximately **■4.86** million, and uncovers elevated cancellation risk around holiday-adjacent bookings.

Section 1 - Data Quality Checks

B1: Multiple property names appeared across different cities and property IDs, confirming that `property_name` alone is not a reliable grouping key. B2: The coupon column used two encodings for no coupon: blank string and 'NONE'. Real coupon usage was calculated after accounting for both representations. B3: 50 cancelled bookings contained review ratings, which is logically impossible because cancelled stays should not generate guest reviews. Recommended treatment: flag and exclude these reviews from rating analysis.

Additional audit findings:

- Invalid stays (checkout <= checkin): 120
- Booking before signup date: 163
- Zero-room bookings: 60
- Loyalty tier parsing issue validated ('None' is a legitimate value)
- Review scale mismatch identified between corporate and non-corporate customers.

Section 2 - Discount Erosion Problem

Platform discount intensity was 5.33%. OTA exhibited the highest discount intensity at 7.30%, which is 1.97 percentage points above the platform average. Customer-mix analysis showed OTA users are somewhat more individual-traveler heavy, but loyalty and city distributions closely matched the platform, suggesting the issue is channel-driven rather than customer-driven.

Coupon effectiveness analysis:

- Cancellation Rate: 22.28% (No Coupon) vs 21.67% (Coupon)
- Per-Room Amount: **■23,209.55** (No Coupon) vs **■20,086.52** (Coupon)
- Repeat Rate: 76.10% (No Coupon) vs 73.99% (Coupon)

Conclusion:

Coupons provide minimal cancellation improvement while reducing basket value and repeat behavior.

Recommended strategy: reduce OTA coupon spend by 50%, recovering approximately **■4.86** million in margin.

Monitor OTA completed booking volume and abort rollback if bookings decline by more than 5% for two consecutive months.

Section 3 - SQL Challenge

A normalized schema was designed with four tables: customers, properties, bookings, and reviews. Primary and foreign keys were defined to preserve referential integrity. A CHECK constraint (checkout_date > checkin_date) was included to address invalid stay records. booking_date was selected for indexing due to frequent time-series analytical queries.

B-Q1:

DENSE_RANK() was used to identify the most-booked room type within each property type.

DENSE_RANK is

preferred because tied room types receive the same rank without gaps, unlike RANK(), and avoids arbitrary

selection unlike ROW_NUMBER().

B-Q2:

Monthly realized revenue for 2024 was calculated using a running cumulative total via window functions.

Full-Year Cumulative Realized Revenue: ■294,859,791.61.

Section 4 - Holiday Proximity Demand Tagger

A holiday enrichment workflow was built using the Nager.Date Public Holidays API. Bookings were tagged

as holiday-adjacent when check-in occurred within ± 2 days of a public holiday.

Results:

- Regular Bookings: 10,994
- Holiday Adjacent Bookings: 1,006
- Avg Booking Value: ■31,327.68 vs ■31,158.23
- Avg Stay Length: 2.85 vs 2.90 nights
- Cancellation Rate: 18.91% vs 22.17%

Non-Obvious Insight:

Holiday-adjacent bookings do not generate materially higher booking value, yet they exhibit significantly

higher cancellation rates. This suggests that peak-period demand is not necessarily more reliable and may

benefit from stricter cancellation policies or advance-payment requirements.

AI Usage:

AI was used to accelerate API integration structure, error handling design, feature engineering ideas,

and visualization planning. All outputs were reviewed, validated, and adapted to the business context

before inclusion in the final analysis.